# **Introducing Grid 2.0:** An open standard for grid service access and allocation

Ensuring affordable electricity for U.S. communities through rapid time-to-power for data centers, large loads, and related resources

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*AI generated image*

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*The views in this paper are not necessarily those of the contributors or the supporting organizations.*

*"Introducing Grid 2.0" was developed by a small group of individuals with deep electric grid and market development experience across multiple organizations. As of publishing, they have formed a working group called the Grid 2.0 Task Force ("G2TF"), which will maintain its work and share progress at g2tf.org.*

*The G2TF was inspired by the Internet Engineering Task Force ("IETF") and its predecessors. Working from the late 1960s to the 1980s and up to today, a small group of motivated industry leaders and researchers worked collectively to solve telecommunications problems beyond organizational boundaries and within existing regulations. Their work literally created the Internet, the rapidly growing, resilient system that grows without central planning and connects billions of people.*

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## Summary and Benefits

Amid mounting concerns over utility ratepayer affordability, the U.S. power sector is experiencing an unprecedented surge in load growth. Historically, grid planning has relied on a conservative strategy of overbuilding infrastructure to accommodate peak demand periods that account for as little as 0.3% of the time, just 30 out of 8760 hours per year. To break away from this costly cycle, the electricity industry must decouple system access from traditional reliability planning while automatically protecting the grid.

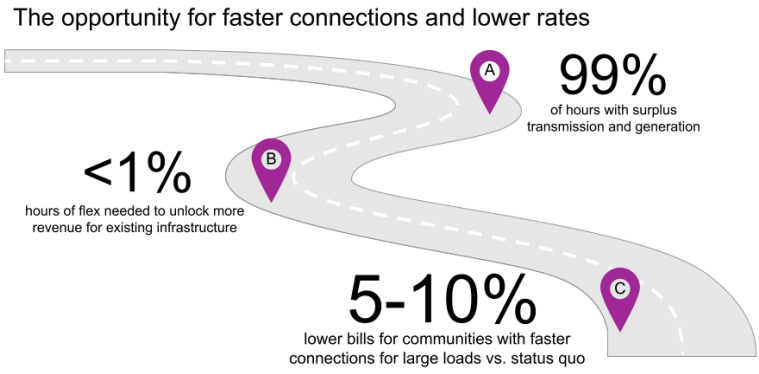



**Figure 1.** Summary view informed by studies from Duke, Purdue, Brattle, and Camus (see "Industry Analysis Informing Grid 2.0")

FERC's recent "show cause" orders[1] for U.S. grid system operators call for aggressive action on interconnection processes and the integration of flexibility for large loads. Promising mechanisms are beginning to surface within multiple ISO/RTO proceedings, such as PJM's Interim Resource Adequacy Service[2] (IRAS), SPP's Conditional High Impact Large Loads[3] (CHILLS), and ERCOT's Provisional Controllable Load Resources[4] (PCLR), but these options remain fragmented, somewhat manual, and embedded in long planning cycles. Unifying these emerging structures and accelerating their deployment will require a voluntary, automated, and standardized approach to implementation.

To this end, a novel Grid 2.0 protocol standard ("Grid 2") is proposed whereby new large loads, transmission, batteries, and clean energy assets voluntarily adopt orchestrated management of their grid participation in a real-time, permission-based paradigm. When implemented by the current queue of grid additions, it will produce an unprecedented level of electric grid resiliency and efficiency through its grid protective, system-wide response.

Grid 2 offers a standard to operationalize and automate the concepts expressed in a growing body of research and bespoke flexible interconnection structures. When implemented as a load-focused operational layer of these rapidly developing market structures, Grid 2 becomes an accelerant, fully recognized in economic dispatch and unit commitment paradigms. All Grid 2 participating assets will continue to meet all NERC, FERC, utility, and TSO/ISO requirements in their current grid context ("Grid 1").

[1] FERC Show Cause Order
[2] PJM IRAS
[3] SPP CHILLS
[4] ERCOT PCLR

This standard addresses several key challenges in the AI, clean energy, and utility contexts simultaneously:

1. **Separating grid access from planning:** By codifying “connect and manage” methods into a single dependable protocol standard, Grid 2 allows planners to separate interconnection permits of large load (and renewables supply) sites from critical peak planning, leading to higher utilization of existing infrastructure.**Fair and efficient allocation:** Grid 2 brings proven modern flow control, scheduling techniques used in internet protocols, like TCP/IP, and cloud and datacenter resource management techniques, such as Kubernetes[5], to the management of scarce grid resources. This low-cost digital architecture overlay is only recently possible for the grid due to the scaling of battery storage, large inverter-based supply resources, and the dynamic dispatch capabilities of some modern datacenters, industrial loads, and smart devices.
2. **Unifying disparate approaches:** By establishing a voluntary digital protocol, Grid 2 brings together disparate systems and stakeholders all struggling with similar issues in their local grid contexts without the need for multiple stakeholder consensus processes leading to a patchwork of solutions in each utility service territory, U.S state, or grid region. A key design feature of Grid 2 is that it does not need to be universally applied. It serves and sits inside the Grid 1 context at any scale or scope.

## Comparison to Current Grid Planning and Operations

Current Grid 1 planning paradigms rooted in building to peak demand and peak supply from renewables, along with slow interconnection processes have failed to keep up with the desired load growth. Speed to power, reliability, and affordability are suffering as a result. This current approach is systematically limiting access to existing underutilized transmission and generation for customers and the associated revenues for utilities due to planning assumptions driven mostly by impacts expected in <1% of hours (see “Industry Analysis Informing Grid 2.0” section).

| Grid 1 - Patchwork of Load Flexibility | Grid 2 - Embedded Load Flexibility |
|---|---|
| • Guaranteed access to power at time of interconnection for life of contract<br>• Achieves reliability via expensive, underutilized buildout of grid and generation assets to new peak with each addition of load<br>• Flexibility is implemented via curtailment of large loads in one-off contracts, operationalized via inefficient hours to | • Permission based access to grid at every time interval with tiered classes of service reflecting different levels of flexibility tolerance, systematically validated by transmission owners and regional grid operators<br>• Guarantees reliability by automatically and proportionally orchestrating participating loads to ensure grid constraints are never |

[5] Kubernetes

| Grid 1 - Patchwork of Load Flexibility | Grid 2 - Embedded Load Flexibility |
|---|---|
| day-ahead warnings when constraints are forecasted | breached<br>• Need to reduce load is limited by buffering storage assets operating according to the same protocol |

Grid 2 offers a new approach: a standardized, automated way to integrate load and supply with differentiated levels of service flexibility. This platform brings the dynamic visibility and control seen on distribution-based demand response implementations to transmission-level loads connections. Network-directed storage-based buffers, behind-the-meter or front-of-meter, reduce the need to call on flexibility by adding it from new, rapidly-connected assets. Grid 2 creates a protocol stack to unify emerging flexibility proposals from utilities and RTOs while setting the foundation to better embrace storage and renewables.

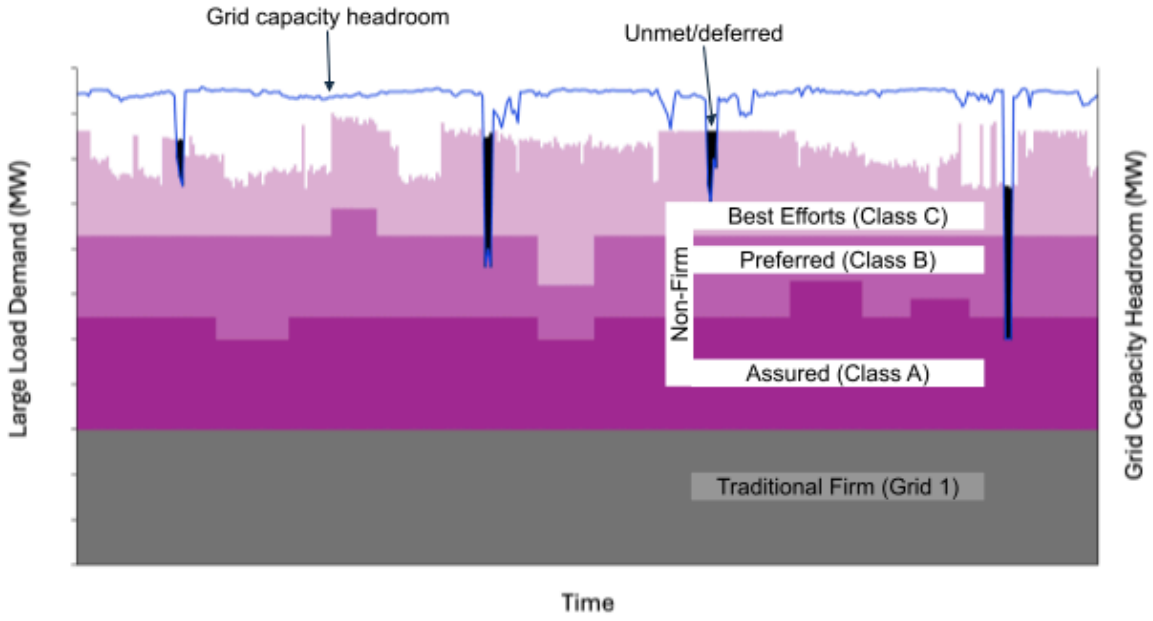



**Figure 2.** Conceptual representation of stratified flexible loads and grid capacity headroom utilization. Classes A, B, and C together represent traditional non-firm demand.

## Proposed Scope

The Grid 2 protocol runs in near real-time dispatch windows of one minute with a goal of managing critical peak congestion in a distributed collective fashion. In initial deployments, it will serve as an operational layer to realize the verifiable dispatch capabilities expected in ERCOT PCLR and PJM IRAS/PRD implementations where parties within those regions are

responsible for flex compliance in the minute(s) timeframe. At launch, version 26.0 of the protocol is focused on units of energy consumed and provided within the time window. Future versions will incorporate other goal-seeks, such as voltage support.

Grid 2 is designed to sit above the sub-second grid operations domain and below the typical five-minute power market clearing domains of Grid 1. With this temporal design respecting Grid 1's existing time scales, Grid 2 can be adopted rapidly on a voluntary basis by select participants in a Grid 2 allocation domain. Stakeholders in ERCOT and PJM are already identifying paths for these control schemes to coexist and reinforce each other. Grid 2 codifies advances in smart C&I demand response into an open industry-wide standard, building on work to-date on VPP and demand response programs across utilities and power markets.

Large loads like datacenters, grid batteries, and renewable assets on the bulk power system will be first to adopt Grid 2 to ensure they can collectively flex around critical peak congestion they would otherwise exacerbate in interconnection planning - leading to delays and higher costs. Hyperscalers are connecting to transmission owners in multiple phases while they wait for firm service to be expanded, creating the opportunity for Grid 2 non-firm access to help accelerate their datacenter site build-outs. Implemented in this way, the increase in non-firm revenues also helps drive down electricity rates for all customers.

In the Grid 2 paradigm, storage-based buffers can be added over time to mitigate periods of unmet demand for participating loads. The same protocol that protects the grid on the load side can ensure protection around Grid 2 compliant storage and renewables supply, creating a more elegant approach to curtailment while supporting rapid interconnections.

As Grid 2 establishes this new "best efforts" electricity paradigm with internet-like statistical multiplexing, it will be generally useful where flexibility can be offered by participants. We expect that equipment of all sizes could adopt the standard and receive benefits in utility programs as a result. Manufacturers of thermostats, refrigerators, and EV chargers could support Grid 2 the way laptop and smart phone hardware and software support TCP/IP - a single orchestrated standard that operates at any scale. Initial work is underway in several markets through the IEEE 2030.5[6] standard, setting a foundation for such a future. That said, Grid 2 is initially focused on the bulk power system implementation and associated benefits at launch with a clear goal of ensuring electricity affordability and rapid interconnections.

[6] IEEE 2030.5 Smart Energy Profile Application Protocol

# The Grid 2.0 Proposal

## Description

The Grid 2 standard consists of a collection of Member Elements and Allocators representing real grid components (load, supply, storage, and wires) all orchestrated on a common protocol stack to schedule equitable permission-based use of (mostly surplus) grid transmission and power capacity enabling rapid grid connections and higher system utilization. Equitable dispatch is proposed as proportional to connection size and differential request size, mirroring the flow control fundamentals of other constrained networks. The standard is designed to span from local transmission congestion zones, to utility and/or balancing authority contexts, and into regional RTOs/ISOs. It can be adopted initially in a local context and extended automatically over time into an open interconnection-wide common use system.

## Structure

In its simplest form, Grid 2 adds a voluntary orchestration plane to the existing grid. This new digital overlay is designed to allocate grid capacity headroom to Grid 2 Members in a dynamic, permission-based fashion. Requests for use of the grid are updated each minute and can be made with tiered levels of priority, the lowest levels of which may be unmet or deferred more frequently.

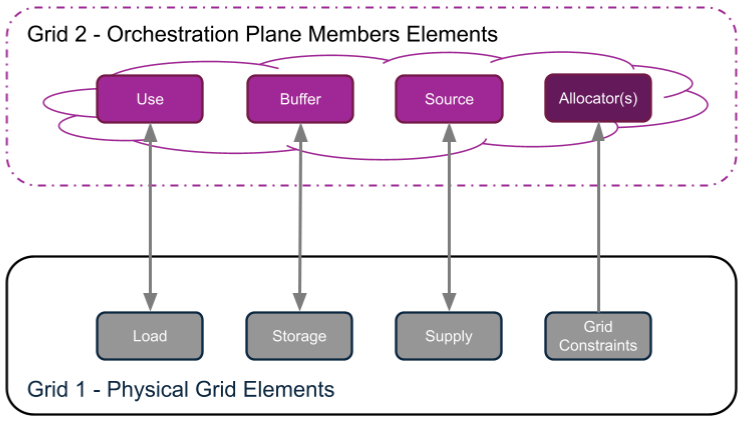



**Figure 3.** Interaction and information flow between Grid 2 orchestration plane and Grid 1 physical plane.

Member Elements each broadcast their differential demand requests (buckets of load classed by priority), production intent, and buffer capabilities. That information is aggregated through Allocators in the local context where it is further informed by Grid 1 capabilities and constraints. The differential service requests are aggregated and cleared each minute within the Grid 1 constraints context by Allocators, which then allocate percentage dispatch for each class of service back to the Member Elements proportional to their requested amounts. Each Member Element then self dispatches its allocated service requests within parameters agreed with their host utility.

A protocol standard is a common way of interacting that allows for wide-spread interoperability and rapid adoption between the components of a system. The orchestration plane of Grid 2 is implemented through three open protocols for grid service access and allocation: Grid 2.0 Protocol (“G2P”), Grid Capacity Allocation Protocol (“GCAP”), and Differential Electric Service Protocol (“DESP”). These are defined in the following sections from various perspectives and will continue to evolve as the Grid 2 standard incorporates feedback from stakeholders.

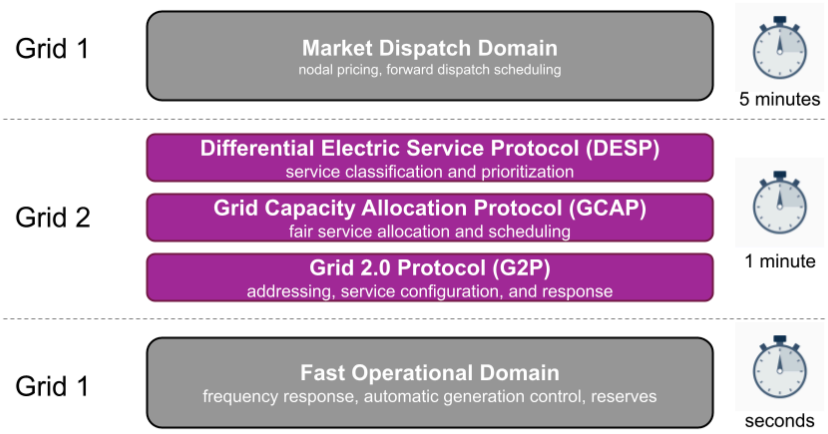



**Figure 4.** Grid 2 protocol stack and market position

## Protocol Interactions

| Member Type | Associated Physical Asset | Broadcasts | Listens for | Intelligence | Failsafe behaviors |
|---|---|---|---|---|---|
| **Use** (load) | Flexible large load (capable datacenter sites[7], electrified industry) | Per-class **request to consume** (+), one signed position per service class (Firm / Semi-Firm / Flex), over a forward time vector *(via G2P)* | Its Allocator's per-class commit-% | **At the edge** — how to map internal workloads onto service classes; which demand is deferrable; anticipating its own forward load | If loses peers/context → reverts to Grid 1 baseline (firm load limit) or manual curtailment; follows allowed ramp rates |
| **Buffer** (storage) | Grid battery / storage asset — the buffer | **Offer to be a source or use** (− discharge / + charge); sits on *either side* of the ledger *(via G2P)* | Its Allocator's committed dispatch — cleared before any load is curtailed | **At the edge** — when to discharge vs. charge for a later peak, when to charge on surplus, protecting a coupled asset; its job is to reduce how often Grid 2 load is unmet | Reverts to its standalone Grid 1 schedule; no Grid 2 support commitment; follows allowed ramp rates |

[7] Can be any combination of flex capable load and/or on-site power resources

| Member Type | Associated Physical Asset | Broadcasts | Listens for | Intelligence | Failsafe behaviors |
|---|---|---|---|---|---|
| **Source** (supply) | Clean-energy / inverter-based supply (solar, wind) | **Offer to supply** (−), generation available this window over a forward time vector *(via G2P)* | Its Allocator's committed take | **At the edge** — forecasting available output; anticipating ramps | Reverts to Grid 1 baseline interconnection behavior; follows allowed ramp rates |
| **Allocator** *(T-Line → Utility load zone → RTO)* | A zone's Grid-1 power production capacity + transmission constraints; can include power distribution factor, dynamics that might be available from Dynamic Line Ratings, etc. | Per-class commit-% back to its Members; federation announcements to peer Allocators *(via G2P)* | Member Service Descriptors in-zone; peer-Allocator announcements across coupled zones | **None internal** — alleviates constraints by redispatching deterministically & proportionally based on a. headroom b. buffers c. ramp down tiered requests (class ordering per DESP) | Stale/partitioned → no clearing; Members fall back to Grid-1 baseline; total pool sized to manageable ramp rates |

All protocol interactions follow two patterns: Member <> Allocator interactions, and Allocator <> Allocator interactions. In Member <> Allocator interactions, Members advertise their requests, Allocators clear them, and Members are responsible for verified self-dispatch. In Allocator <> Allocator interactions, Allocators follow an announce-and-listen federation, setting allocation decisions from information provided by Grid 1 custodians (zone capacity, target reserves, etc.).

In the event of faults between Members and Allocators, and intra-Allocator communication, Member Elements fall back to a Grid 1 failsafe status (what otherwise might have happened under prior curtailment regimes and within aggregate system ramp limits). Grid 2 communications and systems will be built with industry best practices for cyber security and are expected to be implemented in private networks for transmission level assets.

## Key Design Principles and Characteristics

Grid 2.0 was designed with proven principles from the world's highest-growth infrastructures, the Internet and data centers, which deliver higher reliability and higher utilization than the grid with less central planning and regulation. The following table highlights key objectives and rationale for Grid 2 design choices.

| Objective | Approach | Rationale |
|---|---|---|
| **Coexist with physical grid** | Overlay network architecture | Members are embedded throughout Grid 1, not topologically separate; verified by aggregate footprint |
| **Coordination without prices** | Protocol-based coordination | Technical protocols handle real-time coordination; maintains current commercial settlements decoupled on slower clock |
| **Scale without singular operator** | Edge intelligence at every Member Element; extensible design allows for organic growth | Every asset runs the same algorithm against the same observable signals; collective behavior emerges |
| **No surprises** | Failsafe defaults with status quo Grid 1 | Member Element  losing context or peers reverts to baseline behavior; managed to local and regional ramp limits |
| **Bet on current grid physics** | Grid 2 commits only above timescales where Grid 1's primary controls work | Grid 1's ancillaries absorb errors without modification; NERC CPS1/BAL[8] are measures; size of ancillary services can be adjusted up or down over time to match need |

## Utility Level Implementation

The following diagram illustrates a simple utility level implementation of a Grid 2 Allocation Domain.

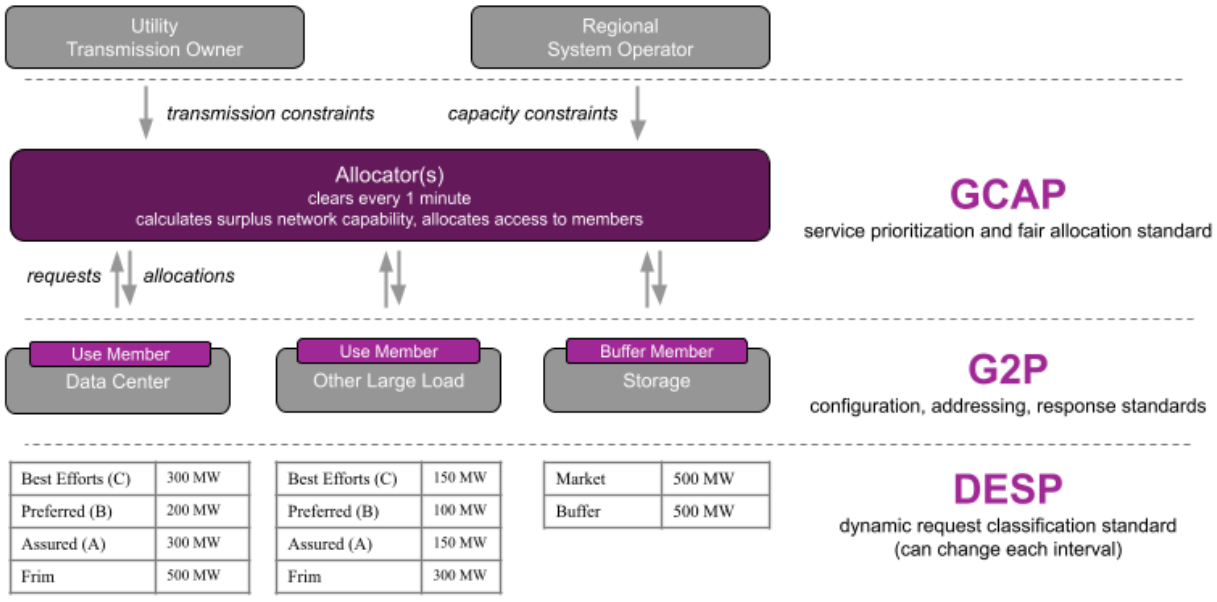



**Figure 5.** Example Transmission Operator implementation with a single constraint domain

[8] NERC BAL Reliability Standard

## Grid 2.0 Membership and Basis of Authority

Numerous announcements have been made about arrangements between utilities and flexible large loads on how they are implementing “connect and manage” arrangements. The Grid 2 design assumes willingness of parties to agree on these principles. With the same or less willingness to flex and accept flex, Grid 2 will produce better outcomes. Therefore, membership in Grid 2 Allocation Domains is expected to be voluntary and can exist distinct from other grid and market requirements, just as emerging “connect and manage” schemes have.

Authority for Grid 2 is anchored in the utility transmission owner where inputs must be provided to the Allocators to define the local and regional constraints for Members Elements in the same domain. Therefore, the smallest Grid 2 Allocation Domain would include a host utility and a flexible load customer.

## Extensible System-Wide Architecture

Grid 2.0 has been designed with an extensible system architecture. It is possible to implement it at a local level and then expand or federate a system with regional sister systems for greater collective benefit (see diagram below).

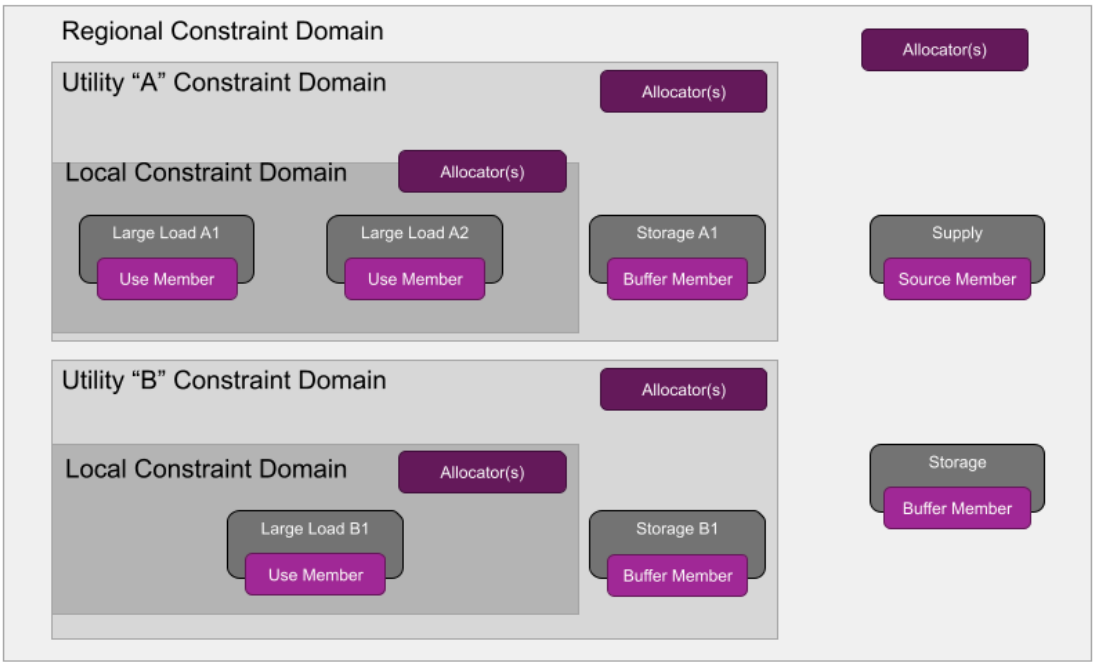



**Figure 6.** Example federated regional implementation across multiple constraint domains

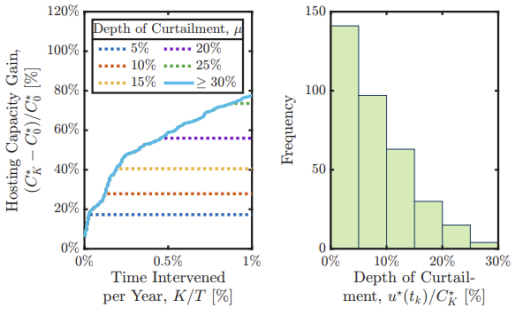
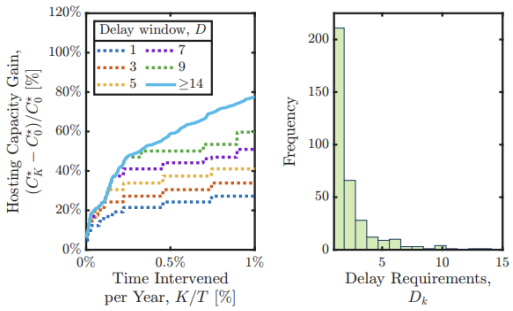
# Industry Analysis Informing Grid 2.0

A significant body of research and analysis is emerging to support the viability of a grid with large load flexibility to facilitate rapid interconnection, enhanced resilience, and improved

affordability. Analysis overwhelmingly concludes that relatively minimal levels of curtailment and flexibility can unlock significant capacity within existing infrastructure, enabling faster time to power, for both loads and new generation. These analyses underpin the core value proposition of the Grid 2.0 standard proposed in this paper.

However, these works generally focus on either data center project level design and interconnection proposals, or high level system capacity potential. These prior studies do not propose a unifying and automated protocol to facilitate a standardized approach to rapid interconnection, a resilient grid, and efficient planning processes, hence we propose Grid 2 as an accelerant for these objectives. Key conclusions from a select subset of studies are highlighted in the table below:

| Study | Authors | Key Findings |
|---|---|---|
| Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in the US Power System (paper) | Norris et al. Duke University, February 2025 | Minimal load flexibility can unlock significant system capacity within the existing grid: 0.25% curtailment can unlock 76GW of new load nationally, while 1.0% curtailment can unlock 126 GW of new capacity. |
| The Untapped Grid: How Better Utilization of the Power System Can Improve Energy Affordability (report) | Hledik et al. The Brattle Group, March 2026 | Large load flexibility can materially increase system utilization, which in turn can lead to substantial rate reductions by 2030. |
| EPRI DCFlex: Data Center Flexibility Framework Initiative (library) | Electric Power Research Institute, Launched October 2024 | A significant body of research and publications supporting the value of flexibility in facilitating faster timer to power and greater grid utilization, which establishes the idea of headroom to drive flex decisions. |
| The Role of Flexible Connection in Accelerating Load Interconnection in Distribution Networks (paper) | Gu et al. Purdue University, October 2025 | Modest flexibility, including curtailment and delay flexibility, can unlock material system capacity. Curtailment interventions during 1% of the year can increase additional system hosting capacity by 80% with interventions almost overwhelmingly below 20% depth of curtailment. Similarly, delay interventions also significantly increased hosting capacity, with short delay windows, less than 1% of the year adding 80% to hosting capacity. |

| Study | Authors | Key Findings |
|---|---|---|
| | |  <br>**Figure 1. Hosting capacity gain for CF (left) and depth of curtailment requirement distribution with $K/T = 1\%$ (right).**<br>  <br>**Figure 4. Hosting capacity gain for DF (left) and delay requirement distribution with $K/T = 1\%$ (right).** |
| Flexible Data Centers: A Faster, More Affordable Path to Power (paper) | Camus, Encoord, Princeton University Zero Lab, December 2025 | Connect now and operate flexibly during grid constraint framework can facilitate time to power for new large loads in 1-2 years. Bring your own capacity ("BYOC") resources on site can be dispatched 40-70 hours per year; a reliable grid remains available over 99% of the year. |
| Powering Reliability Through Market Design (paper) | PJM, May 2026 | Differential Reliability is a possible path to ration scarce generation capacity by geographic region (state) and/or customer class (residential / large load). |
| How we're making data centers more flexible to benefit power grids (blog) | Google, 2025 | Flexible data centers can facilitate better utilization of the grid. Google is implementing demand response deals with utilities. |

In addition to academic analysis, improved utilization of distribution level infrastructure has recently been a prominent focus in Europe and Australia. Concepts such as flexible (interruptible, phased, time-limited) connections, dimming of devices, active network management of solar, wind, batteries, and EV charging hubs, and batteries as congestion

relievers are being implemented. These concepts are similar to emerging proposals on the transmission side, and the market approaches are equally diverse and complex to implement. Grid 2 introduces a protocol to scale these concepts with a common approach.

## Open Questions

While the reliability, speed, and affordability benefits clearly motivate action on the Grid 2 standard, questions remain. Select questions highlighted below illustrate critical details to be developed. For most, we believe there are established technical and regulatory safeguards already embedded in Grid 1 processes on which the Grid 2 standard can develop and grow. Through broad stakeholder engagement remaining issues can be effectively addressed. We note that several of these issues are already being hashed out across power markets in the U.S. With this paper, the Grid 2.0 Task Force invites such stakeholder engagement and the requisite technical studies.

Below is an initial list of questions, which will be updated and resolved in successive feedback cycles with stakeholders:

| | |
|---|---|
| Integration with grid planning processes | • How does Grid 2 impact how reserve margin is calculated?<br>• How does this integrate into established IRP processes?<br>• How should Grid 2 connections be incorporated into reliability planning and interconnection processes governed by RTO?<br>• What happens to all-firm loads in the interconnection queue?<br>• Will the ability to flex out traditional peak load contribution, how should Grid 2 enabled use be charged? |
| Commercial terms | • What might commercial terms with Grid 2 customers look like, particularly with regard to enforcement of flexible capabilities?<br>• Should there be guidance or limits on Grid 2 differential service classes?<br>• How might Grid 2 participating organizations choose to solve their direct congestion needs in this new paradigm? If a new battery is needed to support new Grid 2 participating loads, how is the investment decision made?<br>• How are broader rate structures impacted? |
| Ratepayer protections and value | • What processes are necessary to ensure utilization benefits translate to reduced customer rates while improving utility financial performance? |
| Reliability Assurance | • How do we ensure coincidental load shedding does not cause system stability challenges?<br>• How can we be assured that flexibility commitments will in fact materialize at scale? How should incentives be structured to deliver compliance? |

| | |
|---|---|
| Engineering | • Is there sufficient headroom to absorb coincident swings?<br>• Are there cybersecurity or NERC CIP concerns?<br>• Which existing signals from transmission owners are the best to use for inputs? How can these be standardized? |
| Long term value | • How quickly does increased utilization of the current grid run into requirements for material upgrades with extended timelines? |

## Current Industry Approaches

Several approaches to flexible grid planning and rapid interconnection exist. Grid 2 is intended to be a practical, common standard implementation of many of these proposals. That said, many proposals either inhibit growth, fail to solve interconnection pace and reliability issues, or are piecemeal and bespoke solutions. While potentially effective in each region, these varying solutions dilute expert effort while creating added friction for parties across the ecosystem.

| Approach | Examples | Grid 2.0 Task Force View |
|---|---|---|
| Maintain status quo approach | • Extended interconnection studies<br>• Transmission upgrades and buildout | • Does not meet growing demand in timeframe required<br>• Significant pressure on affordability |
| Limit buildout of new load | • Low data center buildout<br>• Limited industrial electrification and EVs | • Limits economic growth, AI progress, electrification and decarbonization potential |
| Rely on energy markets | • Higher peak energy prices and capacity auctions incentivize loads to curtail or encourage capacity buildout | • Data center operators can tolerate energy costs ranging from 10-100 times the typical gas heat rate benchmark of $50/MWh[9]<br>• Inefficient, ill-timed, build cycles are not aligned with ramp in demand |
| Custom rate structures | • Hyperscaler funds demand response agreement with local utilities | • Piecemeal approach requiring bespoke negotiations<br>• Grid 2 enables implementation |
| Remedial Action Schemes (RAS) | • Transmission operations contingency plans | • Doesn't solve peak generation capacity<br>• Causes stark measures to be implemented toward customers<br>• Grid 2 softens impact to customers |
| Interruptible tariffs | • Demand Response | • Not automated,<br>• Doesn't solve transmission capacity constraints<br>• Grid 2 utilizes differential service to implement more elegant approach to diminished service |
| Emerging flexible or expedited interconnection | • SPP CHILLs<br>• PJM IRAS/PRD<br>• ERCOT Batch Zero/PCLR | • Unique to geography or jurisdictional body with risk of divergence<br>• Policy only, non-operational protocol. Requiring |

[9] Royal, 2026

| Approach | Examples | Grid 2.0 Task Force View |
|---|---|---|
| regimes | • MISO LLWG[10] | non-automated implementation processes<br>• Do not necessarily solve local constraints<br>• Grid 2 enables implementation with added scopes and jurisdictions |
| FERC Directives | • 403 directive[11] on large load interconnection<br>• Section 206 order[12] directing reform on large load interconnection | • Policy directives only<br>• Not operational protocols<br>• Will not necessarily result in unified interconnection approach across RTO/ISOs<br>• Grid 2 provides operational protocol to help resolve interconnection challenges and align action across markets |
| Off-grid large loads | • GW Ranch, CloudBurst | • Expensive, less efficient use of capital<br>• Miss the benefit of sharing overbuilt power systems<br>• Many intend to interconnect eventually |

## Phased Adoption

Grid 2.0 adoption is expected to occur in select grid pockets initially and will likely be a collaboration between a hyperscaler, a utility transmission provider, a clean energy provider, and a load serving entity. The following phases are recommended for deployment:

Phase 0 - Shadow Mode: Run the full protocol stack as a dry run; make and track commitments; execute only what would have happened anyway

Phase 1 - Local Vertical Integration: Groups of large loads, transmission owners, and load serving entities build a solution proof in existing rules structures

Phase 2 - Regional Federation: Peer entities running the same spec interoperate; inter-domain advertisements and policy

### Beyond the Load Growth Challenge

We have chosen to focus primarily on load in this launch document due to the urgency of related policy actions underway across the United States. That said, many Grid 2.0 principles and design characteristics apply beyond large load flexibility and the current load growth challenge. Renewables and storage have been challenged with similar planning and interconnection bottlenecks to those now seen on the load side for many years. The Grid 2.0 Task Force plans to publish two companion documents to this one focused on:

1. Network-directed storage operating within the Grid 2.0 standard to deliver anticipatory, dynamic movement of energy in space and time,

[10] MISO LLWG
[11] FERC 403 Directive
[12] FERC 206 Order

2. Expedited renewables interconnections within Grid 2.0 allowing for smarter separation of rapid access and long-term planning.

Beyond the use, buffering, and production of energy outlined in this paper, we could see the routing of electricity through advanced power electronics based solid-state transformers ("SSTs") and advanced HVDC lines. With its extensible design, Grid 2 is ready-made to help deliver the full system benefits of these innovations.

## Development and Engagement Approach

This initial Grid 2 proposal has been developed by a small group of individuals with deep electric grid and market development experience across multiple organizations. As of publishing, they have formed a working group called the Grid 2.0 Task Force ("G2TF"), which will maintain its work and share progress at g2tf.org.

The G2TF was inspired by the Internet Engineering Task Force[13] ("IETF") and its predecessors. Working from the late 1960s to the 1980s and up to today, a small group of motivated industry leaders and researchers worked collectively to solve telecommunications problems beyond organizational boundaries and within existing regulations. Their work literally created the Internet, the rapidly growing, resilient system that grows without central planning and connects billions of people.

### Development Principles

The need for speed and solutions for a more affordable grid demands a similar approach to rearchitecting the grid. The G2TF will work with an ethos similar to the IETF, consistent with the following principles:

- Start small, stay focused
- Open participation, individual merit
- Time-limited milestones
- Get rough consensus, publish, then run

To this day, a hallmark of the IETF process is the Request for Comments ("RFC") model for Internet standards. In that spirit, this document is G2TF RFC #1, so please provide comments and share with others who might be contributors. If you've made it this far, please consider joining in the work of the G2TF.

### Focused Stakeholder Outreach

The G2TF is actively developing a companion document to this one incorporating detailed regulatory and engineering level concerns. To validate that work, a focused cycle of stakeholder interactions are required. We are starting with the stakeholders with the most pressing issues and those who are calling for solutions. At publishing these include:

[13] About the IETF

1. Sets of hyperscalers, utilities, and state regulators actively working under committed contracts to resolve real grid capacity allocation issues,
2. Researchers and company labs actively modeling these dynamics (EPRI, Purdue University, Duke, Emerald.ai, GridCare, Nvidia),
3. Associated regional transmission and market operators implementing or calling for new approaches (ERCOT, SPP, PJM, MISO),
4. U.S. federal agencies, which are actively supporting and directing progress on these matters (NERC, FERC, DOE).

Grid 2.0 is not intended to be specific to any market or nation. Authors of this paper are actively engaged with the international community through the World Economic Forum, as a way to learn from other markets and share innovations and progress from the U.S.

## Where to Engage

The following engagement tools are planned and will be live before a planned public launch in August 2026.

- www.g2tf.org
- The “G2TF” WhatsApp Community for cross organization real-time news
- g2tf-org Github repositories for open source standards and reference code